

BRIEFING DOCUMENT

Presenting the Knowledge Graph to Dr. Duerfeldt

What to prepare, what to emphasize, and how to connect our Oral Mucositis drug-repurposing engine to his medicinal chemistry workflow.

Who He Is

Adam S. Duerfeldt is an Associate Professor in the Department of Medicinal Chemistry at the University of Minnesota. He holds a Ph.D. from the University of Kansas and did postdoctoral work at Scripps Research Institute (with Dale Boger). Quentin Avila is one of his former doctoral students — our existing connection to the lab.

His lab has two active research programs:

ANTIBACTERIAL

- Small molecules that *activate* bacterial enzymes (not inhibit) — unconventional approach
- Targets: **ClpP** protease, **CpxRA** two-component signaling
- Gram-negative permeation and efflux — quantitative tools for predicting drug penetration
- *C. difficile* antisporeulation

RETINAL DISEASE

- Diabetic retinopathy, AMD, retinopathy of prematurity
- **PPAR α agonists** — SAR studies, prodrug delivery, microparticles
- Alternative to anti-VEGF injections (limited efficacy, high cost, repeated dosing)

His methods include synthetic chemistry, structural biology, and computational approaches. He already thinks computationally. Recent publications include **STING PROTACs**, Gram-negative property-space mapping, and natural product activators of ClpP — all involving systematic structure-activity reasoning of the sort a knowledge graph can accelerate.

What He Needs to Understand About Us

The one-sentence pitch

We built a Neo4j knowledge graph that connects approved drugs, approved

we built a neo4j knowledge graph that connects approved drugs, Ayurvedic phytochemicals, disease-gene associations, and protein targets into a single queryable network — so you can ask "which existing compounds already hit the targets involved in disease X?" and get ranked candidates in minutes instead of months of literature review.

Data sources we integrate

- **ChEMBL** — 3,276 approved drugs with mechanisms, targets, indications, warnings
- **DisGeNET** — gene-disease associations (biomarkers, genetic variants, altered expression)
- **PubChem** — 60,521 chemical-protein interactions
- **IMPPAT** — Indian medicinal plants and phytochemicals
- **MedPlant (BSI)** — 1,915 medicinal plant species
- **TTD / DrugBank** — therapeutic target-drug-gene mappings

What the graph actually connects

Eight node types — Disease, Gene, Drug, Compound, Plant, Formulation, Protein, Therapeutic Area — linked by relationships like TARGETS, TRANSLATES, CONTAINS, TREATS, and ASSOCIATED_WITH. The graph lets you traverse from a disease through its genes to drugs that target those genes, or from an Ayurvedic formulation through its plants and phytochemicals to the protein targets they hit.

What it answers

- Which approved drugs target genes associated with a given disease?
- Can an Ayurvedic formulation be validated by tracing: formulation → plant → phytochemical → protein target → disease gene?
- Which candidates have multi-target coverage (hitting several disease-relevant pathways)?
- How do candidates rank on target count, drug-likeness (Lipinski), and safety profile?

Where Our Project Meets His Workflow

His pipeline (per Quentin): identify candidate molecules → binding studies → narrow to leads → lab testing. Our graph accelerates the first step — surfacing which molecules to put into binding assays in the first place.

CONCRETE OVERLAP POINTS

- **Target identification for his diseases.** We can load DisGeNET associations for diabetic retinopathy, AMD, or *C. difficile* infection and immediately surface which approved drugs or natural products already hit those targets.
- **SAR hypothesis generation.** His PPAR α work involves systematic analog comparison. Our ChEMBL data includes mechanism-of-action and structural classification that could seed SAR explorations.
- **Natural product leads.** His ClpP work already uses natural product activators (sclerotiamide). Our IMPPAT and PubChem data cover 60K+ phytochemical-protein interactions — a much larger search space than manual literature review.
- **Gram-negative permeation.** His property-space mapping work is computational. The graph could complement it by connecting permeation data to target-disease networks.
- **Multi-target formulations.** If an Ayurvedic formulation hits three different targets relevant to retinal inflammation, that's a testable multi-compound hypothesis his lab could evaluate.

Suggested Meeting Agenda

1 Open: gauge his computational comfort

Ask what experience he has with LLMs and computational tools. This calibrates how technical the demo can be. Given his publication on Gram-negative property-space mapping, he's likely comfortable with data-driven approaches.

2 Demo the graph on Oral Mucositis

Walk through a live Neo4j query: disease $\rightarrow$ genes $\rightarrow$ drugs $\rightarrow$ ranked candidates. Show the traversal visually. Then show the Ayurvedic pathway: formulation $\rightarrow$ plant $\rightarrow$ compound $\rightarrow$ target $\rightarrow$ gene $\rightarrow$ disease. The visual graph is the hook — it makes the abstract idea tangible.

3 Pivot to his diseases

"Can we throw your illnesses at it?" Load diabetic retinopathy or *C. difficile* on the spot. Even partial results show the framework's generalizability. This is where he stops seeing a presentation and starts seeing a tool.

4 Ask what's missing

What databases does he wish he had? What targets or pathways would he want integrated?

Bacterial genome data? Permeation assay results? This scopes the collaboration — and signals we're building for his needs, not just showing a finished product.

5 Discuss the iterative loop

The graph generates candidate lists → his lab runs binding studies → results feed back into the graph as new evidence → next round of candidates is better. This is the long-term value proposition: a feedback loop between computational prediction and wet-lab validation.

What to Prepare Before the Meeting

- **Pre-load his diseases.** Run DisGeNET queries for diabetic retinopathy (or "retinal disease") and *Clostridioides difficile* infection before the meeting so the demo isn't cold.
- **Have 2–3 candidate lists ready.** One for OM (our home turf), one for a retinal disease, maybe a partial one for a bacterial target — showing breadth.
- **Know his recent publications.** Be ready to reference the PPAR α SAR work, the ClpP natural product activation, and the Gram-negative permeation mapping. These are natural bridge points.
- **Prepare a one-page visual.** The graph schema (nodes + edges) as a simple diagram. Medicinal chemists think in structures and relationships — a node-edge diagram is native to how they reason.
- **Anticipate the "so what" question.** His lab does synthesis and assays. The answer: the graph shortens the gap between "which molecules should we test" and "here's the ranked list." It doesn't replace his chemistry — it feeds it better starting points.

What to Avoid

- **Don't lead with AI/LLM jargon.** He's a chemist. Lead with the science: targets, mechanisms, disease biology. Mention the AI layer as a future capability, not the headline.
- **Don't oversell the Ayurvedic angle upfront.** His world is synthetic chemistry and approved drugs. Start with ChEMBL-based drug repurposing (familiar territory), then introduce the Ayurvedic pathway as "we also integrate traditional medicine data, validated through the same target network."
- **Don't present it as finished.** It's ~70% through data collection. Frame it as "here's what it can do now, and here's what your input would shape." Collaboration is more

appealing than consumption.

- **Don't ignore the bacteriology gap.** Per Quentin, bacterial targets are a different world. Acknowledge this — "we'd need your guidance on which bacterial databases and target frameworks to integrate."

Compiled from duerfeldtlabs.com (all pages) and OSPF Ayurveda KG project documentation. Contact: aduerfel@umn.edu · 612-624-6184 · 717 Delaware St. SE, Minneapolis, MN 55414.